Category Theory for Mathematical Physics — Exercise Sheet 8

Hopf algebras and quantum groups

Variant: Solutions

Conventions as in Lecture 8: an algebra is (H, μ, ι), a coalgebra $(H, \Delta, \varepsilon_H)$, the antipode S_H; P is the flip; $q \in \mathbb{C}^\times$ is generic, $q = A^2$. On $V_1 = \mathbb{C}^2$, with $e_1 = \uparrow$, $e_2 = \downarrow$, $K = \mathrm{diag}(q, q^{-1})$, $E = e_1 e_2^T$, $F = e_2 e_1^T$, and $\Delta(E) = K \otimes E + E \otimes 1$, $\Delta(F) = 1 \otimes F + F \otimes K^{-1}$, $\Delta(K) = K \otimes K$, $S_H(E) = -K^{-1}E$, $S_H(F) = -FK$, $S_H(K) = K^{-1}$. In this sheet F is that generator, never a functor (a fibre functor in Exercise 5 is written Φ, its structure maps $J_{g,h}$); the module map $V_1 \to V_1^*$ of Exercise 2 is written T. Braid generators carry their index, σ_i. The basis of $V_1 \otimes V_1$ is $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$, and B_q is the matrix of Theorem 8.9. In Exercise 6, *Integrability*'s anisotropy $\cos\eta$ is written Δ_{XXZ}, to keep Δ for the coproduct.

Exercise 1 (Functions and the group algebra). Let G be a finite group.

(i) Show that $\mathbb{C}^G$, with pointwise product, $\Delta(\delta_g) = \sum_{hk=g} \delta_h \otimes \delta_k$, $\varepsilon_H(\delta_g) = \delta_{g,1}$ and $S_H(\delta_g) = \delta_{g^{-1}}$, is a Hopf algebra.

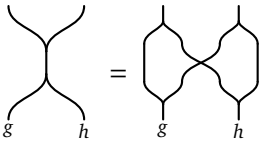
(ii) Show that a $\mathbb{C}^G$-module is the same as a G-graded vector space, that the tensor product of Theorem 8.2 gives $V_g \otimes W_h$ the degree gh, and that the dual of Proposition 8.5 gives $(V_g)^*$ the degree g^{-1}: $\mathrm{Rep}(\mathbb{C}^G)$ is Vec_G.

(iii) Draw the bialgebra law Equation (8.1) for $\mathbb{C}[G]$ on $g \otimes h$, and conclude that $\mathrm{Rep}(\mathbb{C}[G])$ is $\mathrm{Rep}(G)$ with its tensor product.

Solution. (i) $\mathbb{C}^G$ is a commutative algebra with $\delta_a \delta_b = \delta_{a,b}\delta_a$ and unit $1 = \sum_g \delta_g$. Coassociativity: $(\Delta \otimes \mathrm{id})\Delta(\delta_g) = \sum_{xc=g} \sum_{ab=x} \delta_a \otimes \delta_b \otimes \delta_c = \sum_{abc=g} \delta_a \otimes \delta_b \otimes \delta_c$, and $(\mathrm{id} \otimes \Delta)\Delta(\delta_g)$ gives the same sum, by associativity of G. Counit: $(\varepsilon_H \otimes \mathrm{id})\Delta(\delta_g) = \sum_{hk=g} \delta_{h,1}\delta_k = \delta_g$, and likewise on the other side. Δ is an algebra map: $\Delta(\delta_a)\Delta(\delta_b) = \sum_{hk=a} \sum_{h'k'=b} \delta_{h,h'}\delta_{k,k'}\delta_h \otimes \delta_k = \delta_{a,b}\Delta(\delta_a)$, and $\Delta(1) = \sum_{h,k} \delta_h \otimes \delta_k = 1 \otimes 1$; ε_H is evaluation at 1, an algebra map. Antipode: $\mu(S_H \otimes \mathrm{id})\Delta(\delta_g) = \sum_{hk=g} \delta_{h^{-1}}\delta_k$, and $\delta_{h^{-1}}\delta_k \neq 0$ only for $k = h^{-1}$, when $g = hk = 1$; so it is $\delta_{g,1} \sum_k \delta_k = \varepsilon_H(\delta_g)1$. The other side is the same. S_H is invertible (it is its own inverse).

(ii) On a module V the δ_g act by idempotents with $\delta_g\delta_h = 0$ for $g \neq h$ and $\sum_g \delta_g = \mathrm{id}$, so $V = \bigoplus_g V_g$ with $V_g = \delta_g V$; conversely a grading defines such an action. A module map commutes with every δ_g, so it preserves the grading. On $V \otimes W$, δ_k acts by $\Delta(\delta_k) = \sum_{ab=k} \delta_a \otimes \delta_b$, which on $V_g \otimes W_h$ is id if $gh = k$ and 0 otherwise: $V_g \otimes W_h$ has degree gh. The unit $\mathbb{C}$, acted on by ε_H, has degree 1. On V^*, $(\delta_k \cdot f)(v) = f(\delta_{k^{-1}} v)$, so a functional supported on V_g has degree g^{-1}. The associator and unitors are those of Vect, as for Vec_G with trivial ω: $\mathrm{Rep}(\mathbb{C}^G)$ is Vec_G.

(iii) In Equation (8.1) the left side multiplies and then splits; the right side splits both, crosses the middle strands and multiplies in pairs:



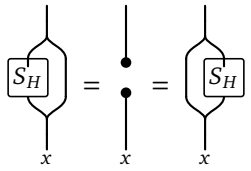
On $g \otimes h$ the left side gives $\Delta(gh) = gh \otimes gh$ and the right side $(\mu \otimes \mu)(g \otimes h \otimes g \otimes h) = gh \otimes gh$. So $\mathbb{C}[G]$ is a bialgebra, and by Theorem 8.2 g acts on $V \otimes W$ by $\rho_V(g) \otimes \rho_W(g)$: the tensor product of $\mathrm{Rep}(G)$. A $\mathbb{C}[G]$-module is a representation of G (extend linearly), and module maps are intertwiners.

Exercise 2 (The antipode of U_q and the dual of V_1). (i) Draw the antipode identities Equation (8.2), with the counit and the unit, and check $\mu(S_H\otimes\mathrm{id})\Delta(x) = \varepsilon_H(x)1 = \mu(\mathrm{id}\otimes S_H)\Delta(x)$ for $x = E, F, K$.

(ii) By Proposition 8.5, x acts on V_1^*, in the dual basis e^1, e^2, by the transpose of $S_H(x)$. Write these matrices, and find the weights of V_1^* (the eigenvalues of K).

(iii) Show that $T\colon V_1 \to V_1^*$, $T(e_1) = e^2$, $T(e_2) = -q\,e^1$, is a module map, and that $(\mathrm{id}\otimes T^{-1})(\mathrm{coev}_{V_1}(1))$ is a multiple of the q-singlet $v = e_1\otimes e_2 - q\,e_2\otimes e_1$.

Solution. (i) The identities, read upward: split x, apply S_H on one leg, multiply; the result is the counit (a strand ending in a dot) followed by the unit (a dot starting a strand), that is $\varepsilon_H(x)1$:



$\Delta(E) = K\otimes E + E\otimes 1$: $\mu(S_H\otimes\mathrm{id})\Delta(E) = S_H(K)E + S_H(E) = K^{-1}E - K^{-1}E = 0$ and $\mu(\mathrm{id}\otimes S_H)\Delta(E) = KS_H(E) + E = -KK^{-1}E + E = 0$, both $\varepsilon_H(E)1$. $\Delta(F) = 1\otimes F + F\otimes K^{-1}$: $S_H(1)F + S_H(F)K^{-1} = F - FKK^{-1} = 0$ and $S_H(F) + FS_H(K^{-1}) = -FK + FK = 0$. $\Delta(K) = K\otimes K$: $S_H(K)K = 1 = KS_H(K)$, which is $\varepsilon_H(K)1$. As S_H is an anti-homomorphism and Δ, ε_H are homomorphisms, the identities on generators give them on all of U_q.

(ii) On V_1, $K^{-1}E = q^{-1}E$ (as $K^{-1}e_1 = q^{-1}e_1$), so $S_H(E) = -q^{-1}e_1e_2^T$ and $E^* = S_H(E)^T = -q^{-1}e_2e_1^T$. Likewise $FK = qF$ (as $e_1^TK = q\,e_1^T$), so $S_H(F) = -q\,e_2e_1^T$ and $F^* = -q\,e_1e_2^T$; and $K^* = (K^{-1})^T = \mathrm{diag}(q^{-1}, q)$. So e^1 has weight q^{-1} and e^2 has weight q: the weights are reversed, as for the dual of a spin.

(iii) $T = \begin{pmatrix}0 & -q\\ 1 & 0\end{pmatrix}$ in the bases e_i, e^i. Then $TK = \begin{pmatrix}0 & -1\\ q & 0\end{pmatrix} = K^*T$, $TE = \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}$ and $E^*T = \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}$, $TF = \begin{pmatrix}-q & 0\\ 0 & 0\end{pmatrix} = F^*T$, so T is a module map, invertible: $V_1 \cong V_1^*$. With $T^{-1}(e^1) = -q^{-1}e_2$ and $T^{-1}(e^2) = e_1$, $(\mathrm{id}\otimes T^{-1})(e_1\otimes e^1 + e_2\otimes e^2) = -q^{-1}e_1\otimes e_2 + e_2\otimes e_1 = -q^{-1}v$. Since coev and T^{-1} are module maps, so is this: it is another proof that the q-singlet is invariant, and the self-duality of V_1 is the cup of Lecture 9.

Exercise 3 (R is not the braiding). On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$, with $\Delta(g) = g\otimes g$, let $R = \frac{1}{2}(1\otimes 1 + 1\otimes g + g\otimes 1 - g\otimes g)$.

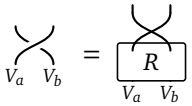
(i) Show that $R^2 = 1\otimes 1$, $(\Delta\otimes\mathrm{id})(R) = R_{13}R_{23}$ and $(\mathrm{id}\otimes\Delta)(R) = R_{13}R_{12}$, and explain why $R\Delta(x)R^{-1} = \Delta^{\mathrm{op}}(x)$ holds.

(ii) On the modules V_a (g acts by $(-1)^a$), show that R acts on $V_a\otimes V_b$ by $(-1)^{ab}$. Draw $c = P\circ R$ as in Equation (8.3), and say which of R and c is a morphism $V\otimes W \to W\otimes V$.

Solution. (i) With $R = \frac{1}{2}\sum_{i,j\in\{0,1\}}(-1)^{ij}g^i\otimes g^j$, $R^2 = \frac{1}{4}\sum_{i,j,k,l}(-1)^{ij+kl}g^{i+k}\otimes g^{j+l}$; the coefficient of $g^m\otimes g^n$ is $\frac{1}{4}\sum_{i,j}(-1)^{ij+(m-i)(n-j)}$, which is 1 for $m = n = 0$ and 0 otherwise. Since $\Delta(g^i) = g^i\otimes g^i$, $(\Delta\otimes\mathrm{id})(R) = \frac{1}{2}\sum(-1)^{ij}g^i\otimes g^i\otimes g^j$, while $R_{13}R_{23} = \frac{1}{4}\sum(-1)^{ij+kl}g^i\otimes g^k\otimes g^{j+l}$; collecting $g^i\otimes g^k\otimes g^n$ gives $\frac{1}{4}\sum_j(-1)^{ij+k(n-j)} = \frac{1}{4}(-1)^{kn}\sum_j(-1)^{j(i-k)}$, which is $\frac{1}{2}(-1)^{in}$ for $k = i$ and 0 for $k\neq i$: the same. The other identity is the mirror computation. Finally $\mathbb{C}[\mathbb{Z}_2]$ is commutative and Δ is cocommutative, $\Delta^{\mathrm{op}} = \Delta$, so $R\Delta(x)R^{-1} = \Delta(x) = \Delta^{\mathrm{op}}(x)$.

(ii) On $V_a\otimes V_b$, $g^i\otimes g^j$ acts by $(-1)^{ia+jb}$, so R acts by $\frac{1}{2}(1 + (-1)^b + (-1)^a - (-1)^{a+b})$, which is 1 unless $a = b = 1$, when it is $\frac{1}{2}(1-1-1-1) = -1$: $(-1)^{ab}$. The picture of $c = P\circ R$ is the

box R on the two strands followed by the plain crossing (Equation (8.3)):



R maps $V\otimes W$ to $V\otimes W$; only $c = P\circ R$ maps $V\otimes W$ to $W\otimes V$ and is the braiding. Here $c(v\otimes w) = (-1)^{ab} w\otimes v$ is the braiding of sVec: $R^2 = 1$ and $c^2 = 1$, yet $R\neq 1$ and $c\neq P$. With $R = 1\otimes 1$ the same Hopf algebra has the symmetric braiding P: the braiding is the choice of R.

Exercise 4 (Hecke, Temperley–Lieb and Clebsch–Gordan). (i) Show that B_q has eigenvalue q on $\uparrow\uparrow$, $\downarrow\downarrow$ and $t = e_1\otimes e_2 + q^{-1}e_2\otimes e_1$, and $-q^{-1}$ on $v = e_1\otimes e_2 - q\,e_2\otimes e_1$; conclude the Hecke relation $(B_q - q)(B_q + q^{-1}) = 0$.

(ii) Show that $U = B_q - q = -q^{-1}vv^T$, that $U^2 = -(q+q^{-1})U$, and that $U_1U_2U_1 = U_1$ on $V_1^{\otimes 3}$ ($U_1 = U\otimes 1$, $U_2 = 1\otimes U$).

(iii) Show that $\Delta(E)v = \Delta(F)v = 0$ and $\Delta(K)v = v$, and that $t = \Delta(F)(e_1\otimes e_1)$. Normalise v and t, and take $q\to 1$: which Clebsch–Gordan states appear?

(iv) With $q = A^2$ and $R = \mathrm{diag}(A, A^{-1}, A^{-1}, A)\big(1 + (q-q^{-1})F\otimes E\big)$ (Theorem 8.9), compute $c = P\circ R$ and show $c = A^{-1}B_q$. What are the eigenvalues of c?

Solution. (i) B_q fixes the lines of $\uparrow\uparrow$ and $\downarrow\downarrow$ with eigenvalue q, and acts on the span of $\uparrow\downarrow, \downarrow\uparrow$ by $\begin{pmatrix} q-q^{-1} & 1\\ 1 & 0\end{pmatrix}$. On $t = (1, q^{-1})$ this gives $(q - q^{-1} + q^{-1}, 1) = (q, 1) = q\,t$; on $v = (1, -q)$ it gives $(q - q^{-1} - q, 1) = (-q^{-1}, 1) = -q^{-1}v$. The four vectors form a basis (t and v are independent since $q^{-1}\neq -q$ for generic q), and $(B_q - q)(B_q + q^{-1})$ kills each of them, so it is 0.

(ii) vv^T is $\begin{pmatrix} 1 & -q\\ -q & q^2\end{pmatrix}$ on the middle block, and $B_q - q = \begin{pmatrix} -q^{-1} & 1\\ 1 & -q\end{pmatrix} = -q^{-1}\begin{pmatrix} 1 & -q\\ -q & q^2\end{pmatrix}$ there, 0 elsewhere: $U = -q^{-1}vv^T$. Then $U^2 = q^{-2}v(v^Tv)v^T = q^{-2}(1+q^2)vv^T = -q^{-1}(1+q^2)\,U = -(q+q^{-1})U$. For the third relation, write $U_1 = -q^{-1}(v\otimes 1)(v^T\otimes 1)$ and $U_2 = -q^{-1}(1\otimes v)(1\otimes v^T)$. Then $U_1U_2U_1 = -q^{-3}(v\otimes 1)\,W_1W_2\,(v^T\otimes 1)$ with the two zig-zags $W_1 = (v^T\otimes 1)(1\otimes v)$ and $W_2 = (1\otimes v^T)(v\otimes 1)$, maps $V_1\to V_1$. With $v_{12} = 1$, $v_{21} = -q$: $W_1e_c = \sum_{a,b} v_{ca}v_{ab}e_b$, which is $v_{12}v_{21}e_1 = -q\,e_1$ for $c = 1$ and $v_{21}v_{12}e_2 = -q\,e_2$ for $c = 2$, so $W_1 = -q$, and likewise $W_2 = -q$. Hence $U_1U_2U_1 = -q^{-3}q^2(vv^T\otimes 1) = U_1$. These are the Temperley–Lieb relations of Lecture 9, with loop value $-(q+q^{-1})$.

(iii) $\Delta(E) = K\otimes E + E\otimes 1$ sends $e_1\otimes e_2$ to $Ke_1\otimes Ee_2 = q\,e_1\otimes e_1$ and $e_2\otimes e_1$ to $Ee_2\otimes e_1 = e_1\otimes e_1$, so $\Delta(E)v = q\,e_1\otimes e_1 - q\,e_1\otimes e_1 = 0$. $\Delta(F) = 1\otimes F + F\otimes K^{-1}$ sends $e_1\otimes e_2$ to $Fe_1\otimes K^{-1}e_2 = q\,e_2\otimes e_2$ and $e_2\otimes e_1$ to $e_2\otimes Fe_1 = e_2\otimes e_2$: $\Delta(F)v = q\,e_2\otimes e_2 - q\,e_2\otimes e_2 = 0$. $\Delta(K) = K\otimes K$ is 1 on $e_1\otimes e_2$ and $e_2\otimes e_1$. And $\Delta(F)(e_1\otimes e_1) = e_1\otimes e_2 + e_2\otimes K^{-1}e_1 = e_1\otimes e_2 + q^{-1}e_2\otimes e_1 = t$: the $m = 0$ state of the q-triplet, lowered from $\uparrow\uparrow$. As $q\to 1$ the normalised vectors become $(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2}$ and $(\uparrow\downarrow + \downarrow\uparrow)/\sqrt{2}$: the singlet and the $m = 0$ triplet state, with the Clebsch–Gordan coefficients $\pm 1/\sqrt{2}$ of *QM-I*.

(iv) $F\otimes E = e_2e_1^T\otimes e_1e_2^T$ sends $\uparrow\downarrow = e_1\otimes e_2$ to $e_2\otimes e_1 = \downarrow\uparrow$ and kills the other three basis vectors: it is the matrix unit in position $(\downarrow\uparrow, \uparrow\downarrow)$. So $1 + (q-q^{-1})F\otimes E$ is the identity with $q - q^{-1}$ in that position, and multiplying by $\mathrm{diag}(A, A^{-1}, A^{-1}, A)$ on the left scales the rows: R has diagonal A, A^{-1}, A^{-1}, A and $A^{-1}(q-q^{-1})$ in position $(\downarrow\uparrow, \uparrow\downarrow)$. P swaps the middle two rows, giving rows $(A,0,0,0)$, $(0, A^{-1}(q-q^{-1}), A^{-1}, 0)$, $(0, A^{-1}, 0, 0)$, $(0,0,0,A)$. With $A = A^{-1}q$ this is $A^{-1}B_q$. Its eigenvalues are $A^{-1}q = A$ (on the triplet) and $A^{-1}(-q^{-1}) = -A^{-3}$ (on v), by (i): $c^2\neq 1$, a braiding that is not a symmetry.

Exercise 5 (No fibre functor). A fibre functor Φ is a faithful $\mathbb{C}$-linear monoidal functor to $\mathrm{Vect}_{\mathrm{fd}}$ (Lecture 8). You may use that a $\mathbb{C}$-linear functor between semisimple categories preserves direct sums.

(i) Show that a fibre functor on Fibonacci would give an integer $n = \dim \Phi(\tau)$ with $n^2 = n + 1$, and conclude that there is none. Do the same for Ising.

(ii) The semion category has every FPdim equal to 1. Show with Sheet 3, Exercise 4, that it has no fibre functor either.

(iii) Why does this not contradict the existence of a fibre functor on Vec_G for $G = \mathbb{Z}_3$ with the pivotal structure $d = e^{2\pi i/3}$ of Lecture 4?

Solution. (i) Φ is monoidal, so $\Phi(1) \cong \mathbb{C}$ and $\Phi(\tau) \otimes \Phi(\tau) \cong \Phi(\tau \otimes \tau) \cong \Phi(1 \oplus \tau) \cong \Phi(1) \oplus \Phi(\tau)$, using that Φ preserves direct sums. Taking dimensions, $n^2 = 1 + n$ with $n = \dim \Phi(\tau)$ a non-negative integer. Its roots are φ and $-1/\varphi$, neither an integer (and $n = 0$ would make Φ kill τ, against faithfulness). So Fibonacci has no fibre functor. For a Hopf algebra H the forgetful functor $\mathrm{Rep}(H) \to \mathrm{Vect}_{\mathrm{fd}}$ is one (it is strict monoidal by Theorem 8.2, and faithful), so Fibonacci is not $\mathrm{Rep}(H)$; Theorem 8.8 gives the converse. For Ising, $\psi \otimes \psi \cong 1$ gives $\dim \Phi(\psi)^2 = 1$, so $\dim \Phi(\psi) = 1$, and $\sigma \otimes \sigma \cong 1 \oplus \psi$ gives $n_\sigma^2 = 2$ for $n_\sigma = \dim \Phi(\sigma)$: no integer.

(ii) In Vec_G^ω on $\mathbb{Z}_2$, $\dim \Phi(s)^2 = \dim \Phi(1) = 1$, so each $\Phi(\delta_g)$ is a line. Choose isomorphisms $\Phi(\delta_g) \cong \mathbb{C}$; transported along them, Φ becomes a monoidal functor that sends each δ_g to $\mathbb{C}$, with structure maps $J_{g,h}$ numbers, as in Sheet 3, Exercise 4(ii). There it was shown that such a functor exists only if $\omega = \delta j$, and in Exercise 4(iii) that $\omega(s, s, s) = -1$ is not a coboundary. So the semion category has none, although its FPdim's are integers: integer dimensions are necessary, not sufficient.

(iii) The obstruction in (i) is FPdim, which a fibre functor turns into ordinary dimensions. The pivotal dimension d depends on a pivotal structure, which Φ need not respect. On $\mathrm{Vec}_{\mathbb{Z}_3}$ every FPdim is 1 and the forgetful functor is a fibre functor, whatever pivotal structure we put on the category; $d = e^{2\pi i/3}$ is no obstruction.

Exercise 6 ⋆. *The XXZ dictionary. Integrability* (its eq. (15.22)) writes the trigonometric R-matrix, in the basis $\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow$, as

$$R(\lambda) = \begin{pmatrix} \sinh(\lambda - i\eta) & & & \\ & \sinh\lambda & -i\sin\eta & \\ & -i\sin\eta & \sinh\lambda & \\ & & & \sinh(\lambda - i\eta) \end{pmatrix},$$

with $\Delta_{\mathrm{XXZ}} = \cos\eta$, η real and generic. Put $q = e^{i\eta}$.

(i) Let $D(\lambda) = \mathrm{diag}(e^{\lambda/2}, e^{-\lambda/2})$ and $\tilde{R}(\lambda) = (D(\lambda) \otimes 1)\, R(\lambda)\, (D(\lambda) \otimes 1)^{-1}$. Compute $\tilde{R}(\lambda)$. You may use that $\tilde{R}$ satisfies the same Yang–Baxter equation as R.

(ii) Show that $\lim_{\lambda\to-\infty}(-2e^{\lambda})\, P\tilde{R}(\lambda) = B_q$ and $\lim_{\lambda\to+\infty} 2e^{-\lambda}\, P\tilde{R}(\lambda) = B_q^{-1}$.

(iii) Show that without the gauge, $\lim_{\lambda\to+\infty} 2e^{i\eta}e^{-\lambda}PR(\lambda)$ satisfies the braid relation but not the Hecke relation.

(iv) Assemble the dictionary: q and Δ_{XXZ}; the over- and under-crossings with their scalars $A^{\mp 1}$; and the point $\eta \to 0$.

Solution. (i) $D \otimes 1 = \mathrm{diag}(e^{\lambda/2}, e^{\lambda/2}, e^{-\lambda/2}, e^{-\lambda/2})$. Conjugating multiplies the entry in row r, column s by d_r/d_s; the diagonal is unchanged, the entry $(\uparrow\downarrow, \downarrow\uparrow)$ is multiplied by $e^{\lambda/2}/e^{-\lambda/2} =$

e^{λ} and $(\downarrow\uparrow, \uparrow\downarrow)$ by $e^{-\lambda}$:

$$\tilde{R}(\lambda) = \begin{pmatrix} \sinh(\lambda - i\eta) & & & \\ & \sinh\lambda & -i\sin\eta\, e^{\lambda} & \\ & -i\sin\eta\, e^{-\lambda} & \sinh\lambda & \\ & & & \sinh(\lambda - i\eta) \end{pmatrix}.$$

(It still satisfies the Yang–Baxter equation, because $R(\lambda)$ commutes with $D \otimes D$, which conserves the total spin; the script checks it on samples.)

(ii) As $\lambda \to -\infty$, $-2e^{\lambda}\sinh(\lambda - i\eta) \to e^{i\eta} = q$ and $-2e^{\lambda}\sinh\lambda \to 1$, while $-2e^{\lambda}(-i\sin\eta\, e^{\lambda}) \to 0$ and $-2e^{\lambda}(-i\sin\eta\, e^{-\lambda}) = 2i\sin\eta = q - q^{-1}$. So $-2e^{\lambda}\tilde{R} \to \begin{pmatrix} q & & & \\ & 1 & 0 & \\ & q - q^{-1} & 1 & \\ & & & q \end{pmatrix}$, and P, which swaps the two middle rows, gives exactly B_q. As $\lambda \to +\infty$, $2e^{-\lambda}\sinh(\lambda - i\eta) \to q^{-1}$, $2e^{-\lambda}\sinh\lambda \to 1$, the entry $(\uparrow\downarrow, \downarrow\uparrow)$ gives $-2i\sin\eta = -(q - q^{-1})$ and the other 0; after P this is $\begin{pmatrix} q^{-1} & & & \\ & 0 & 1 & \\ & 1 & -(q - q^{-1}) & \\ & & & q^{-1} \end{pmatrix}$, which is B_q^{-1}: the middle block of B_q, $\begin{pmatrix} q - q^{-1} & 1 \\ 1 & 0 \end{pmatrix}$, has inverse $\begin{pmatrix} 0 & 1 \\ 1 & -(q - q^{-1}) \end{pmatrix}$.

(iii) Without the gauge both off-diagonal entries stay $-i\sin\eta$: $2e^{i\eta}e^{-\lambda}\sinh(\lambda - i\eta) \to 1$, $2e^{i\eta}e^{-\lambda}\sinh\lambda \to q$, and $2e^{i\eta}e^{-\lambda}(-i\sin\eta) \to 0$. So $2e^{i\eta}e^{-\lambda}R \to \mathrm{diag}(1, q, q, 1)$, and after P the limit is $X = \begin{pmatrix} 1 & & & \\ & 0 & q & \\ & q & 0 & \\ & & & 1 \end{pmatrix}$, a swap twisted by q: $X(e_a \otimes e_b) = q^{[a \neq b]} e_b \otimes e_a$. Both sides of the braid relation send $e_a \otimes e_b \otimes e_c$ to $e_c \otimes e_b \otimes e_a$ times $q^{[a\neq b]+[a\neq c]+[b\neq c]}$, so it holds. But the eigenvalues of X are $1, 1, q, -q$, three distinct numbers for generic q, so no scalar multiple of X satisfies any quadratic relation, while every scalar multiple of B_q (in particular the braiding $A^{-1}B_q$) does: X is not, up to a scalar, the quantum-group braiding. In particular it fails the Hecke relation. The gauge of (i) is what makes the limit B_q.

(iv) $q = e^{i\eta} = A^2$ and $\Delta_{XXZ} = \cos\eta = (q + q^{-1})/2$. The limit $\lambda \to -\infty$ gives B_q, and the braiding of Theorem 8.9 is the over-crossing $c = A^{-1}B_q$; the limit $\lambda \to +\infty$ gives B_q^{-1}, and the under-crossing is $c^{-1} = AB_q^{-1}$. The two ends of the spectral line are the two crossings. At $\eta \to 0$, $q \to 1$ and $B_q \to P$: the rational (XXX) point, where the braid group collapses to the symmetric group. This is the table of § 8.5.